

STM32F070x6

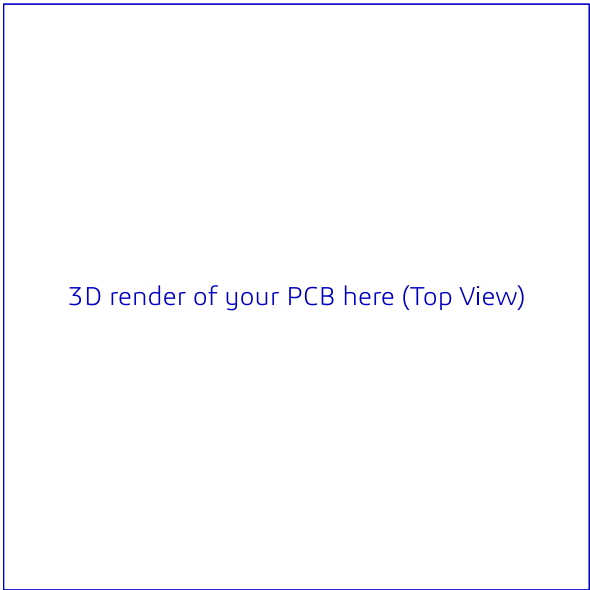
Variant: Variant Name

2025-10-28

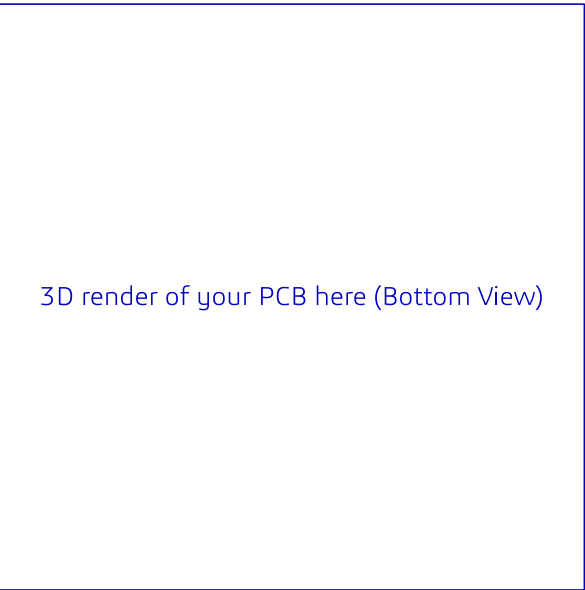
Rev 1.2

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TOP VIEW



BOTTOM VIEW



NOTES

Add a comment here

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Block Diagram

File: block_diagram.kicad_sch

Revision History

File: revision_history.kicad_sch

Project Architecture

File: project_architecture.kicad_sch

Power - Sequencing

File: power_sequencing.kicad_sch

DESIGN CONSIDERATIONS

DESIGN NOTE:

Example text for informational design notes.

DESIGN NOTE:

Example text for debug notes.

DESIGN NOTE:

Example text for cautionary design notes.

DESIGN NOTE:

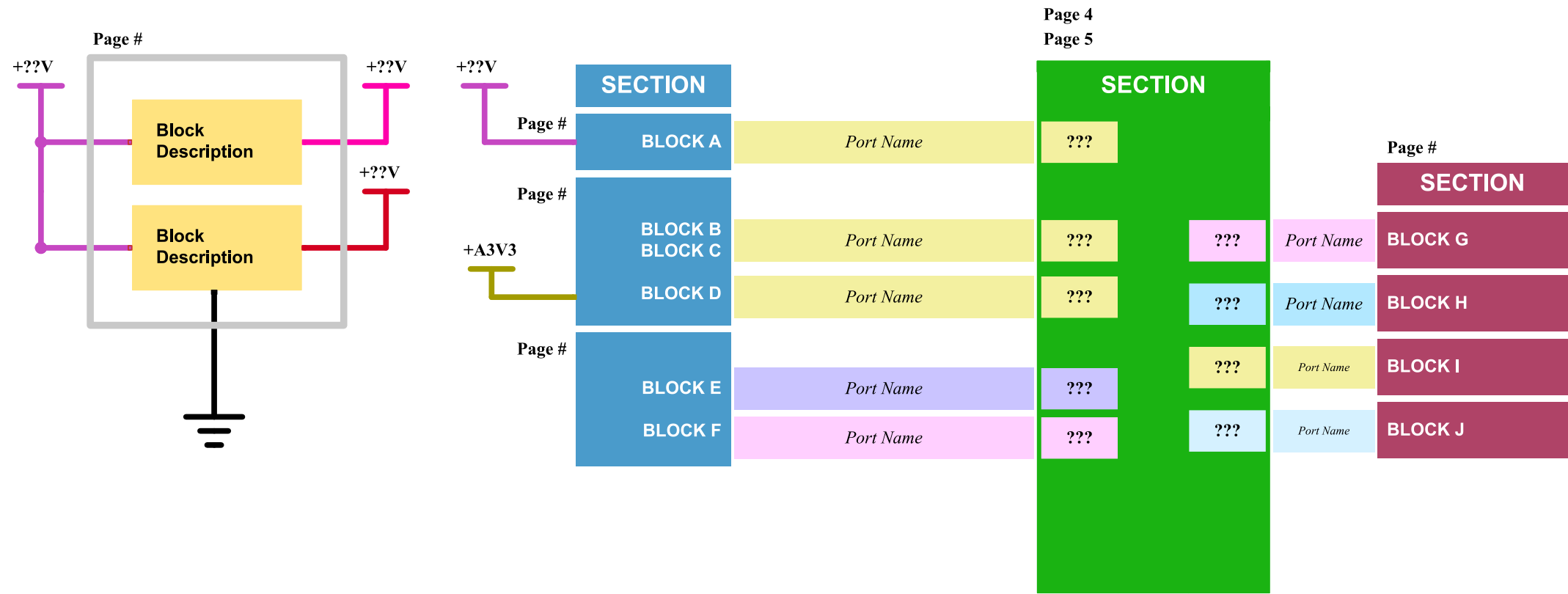
Example text for critical design notes.

LAYOUT NOTE:

Example text for critical layout guidelines.

	Comments:	Company: Ahmad Taka Personal		Variant: Variant Name	
	Sheet Title: Cover Page	Board Name: STM32F070x6		Project Name: Flashlight Project	
	Sheet Path: /	File Name: flashlight-full-v1-1.kicad_sch	Designer: Ahmad Taka	Date: 2025-04-26	Revision: 1.2
			Reviewer: Travis Ziegler	Size: A3	Sheet: 1 of 12

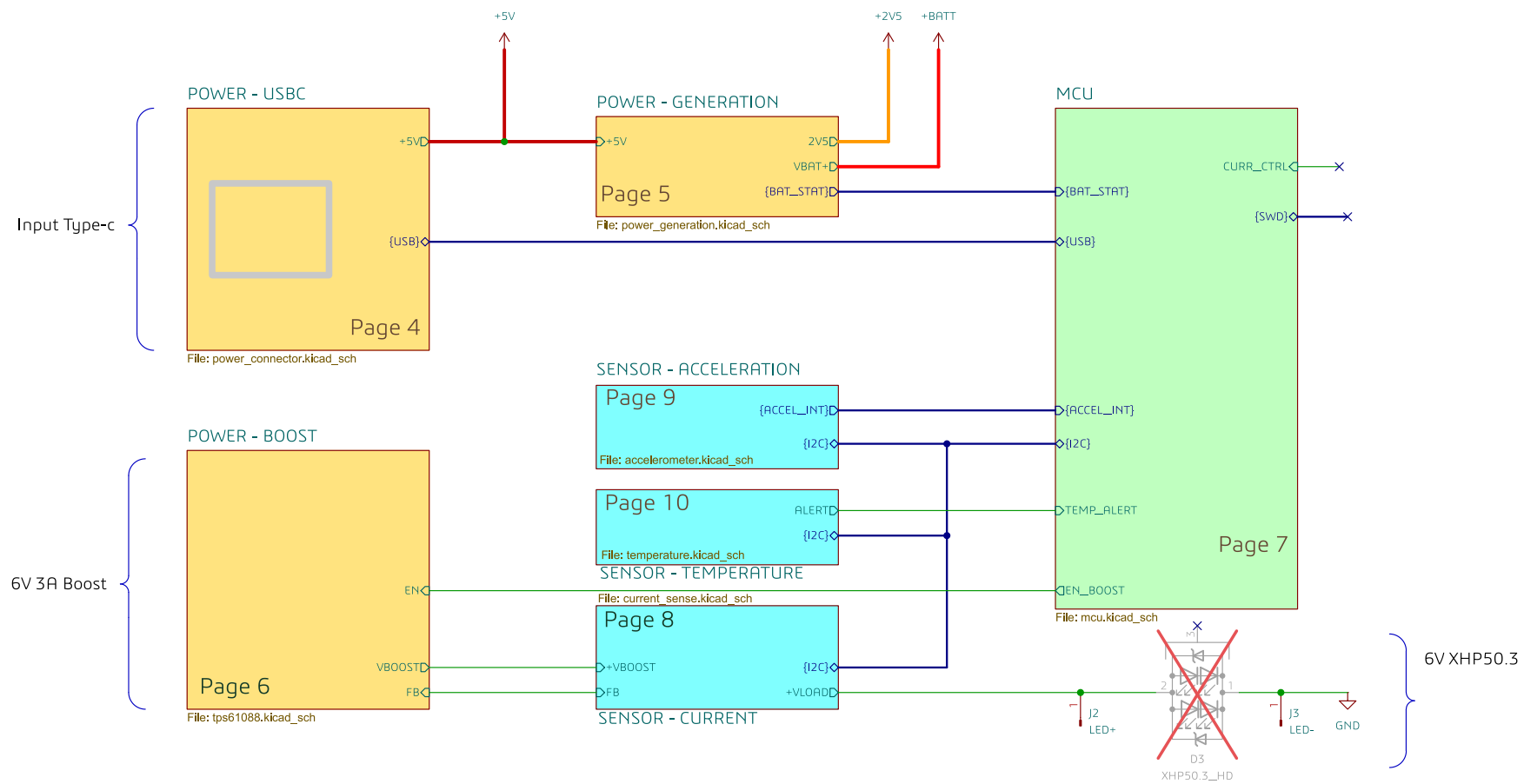
[2] Block Diagram



Target specifications:	
Input voltage:	?? - ?? V
Spec 2	??
Spec 3	??
Spec 4	??

	Comments:	Company: Ahmad Taka Personal		Variant: Variant Name	
		Board Name: STM32F070x6		Project Name: Flashlight Project	
	Sheet Title: Block Diagram	File Name: block_diagram.kicad_sch	Designer: Ahmad Taka	Date: 2025-04-26	Revision: 1.2
Sheet Path: /Block Diagram/			Reviewer: Travis Ziegler	Size: A3	Sheet: 2 of 12

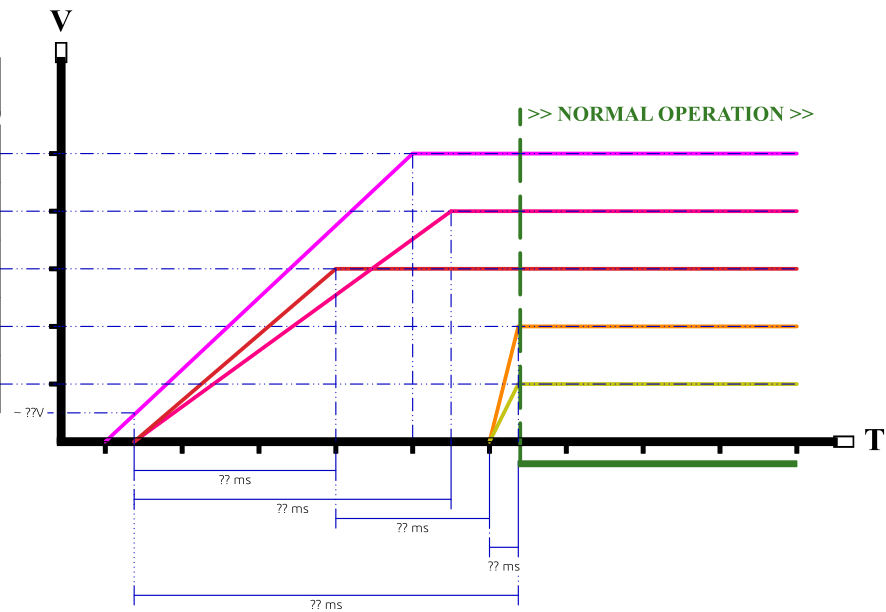
[3] Project Architecture



	Comments:		Company:		Variant:	
			Ahmad Taka Personal		Variant Name	
			Board Name:		Project Name:	
			STM32F070x6		Flashlight Project	
	Sheet Title:	File Name:		Designer:	Date:	Revision:
	Project Architecture	project_architecture.kicad_sch		Ahmad Taka	2025-04-26	1.2
	Sheet Path:	Reviewer:		Size:	Sheet:	
	/Project Architecture/	Travis Ziegler		A4	3 of 12	

[11] Power - Sequencing

NAME	SOURCE	LEVEL
+??V	SOURCE NAME	??V ± ??%
+??V	SOURCE NAME	??V ± ??%
+??V	SOURCE NAME	??V ± ??%
+??V	SOURCE NAME	??V ± ??%
+??V	SOURCE NAME	??V ± ??%



	Comments:	Company: Ahmad Taka Personal		Variant: Variant Name	
		Board Name: STM32F070x6		Project Name: Flashlight Project	
	Sheet Title: Power - Sequencing	File Name: power_sequencing.kicad_sch	Designer: Ahmad Taka	Date: 2025-04-26	Revision: 1.2
	Sheet Path: /Power - Sequencing/		Reviewer: Travis Ziegler	Size: A4	Sheet: 11 of 12

[12] Revision History

DD.MM.YYYY - xxx Revision
Variant: xxx

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	Comments:	Company: Ahmad Taka Personal		Variant: Variant Name	
		Board Name: STM32F070x6		Project Name: Flashlight Project	
	Sheet Title: Revision History	File Name: revision_history.kicad_sch	Designer: Ahmad Taka	Date: 2025-04-26	Revision: 1.2
	Sheet Path: /Revision History/		Reviewer: Travis Ziegler	Size: A4	Sheet: 12 of 12

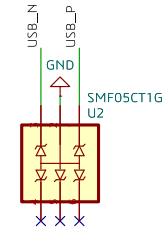
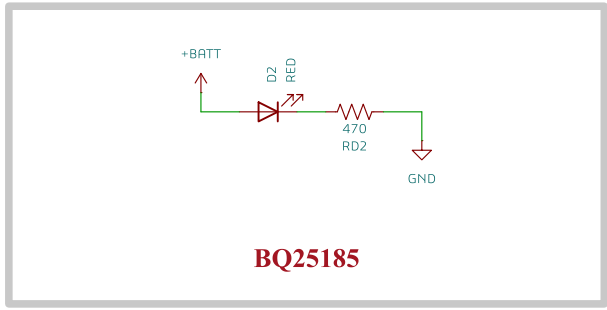
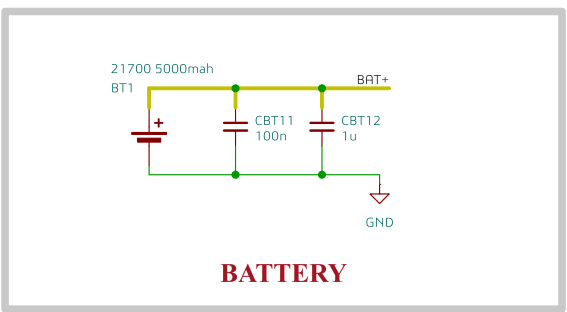
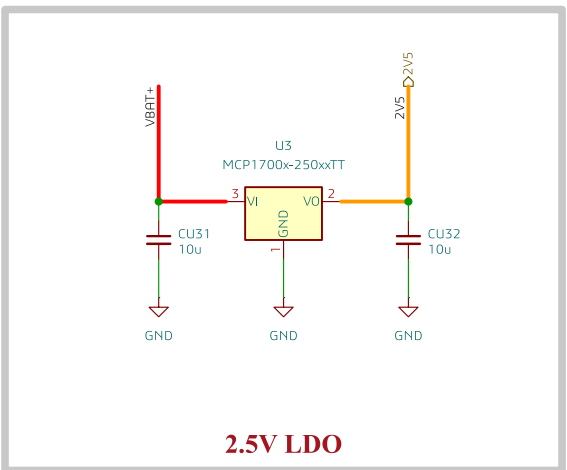
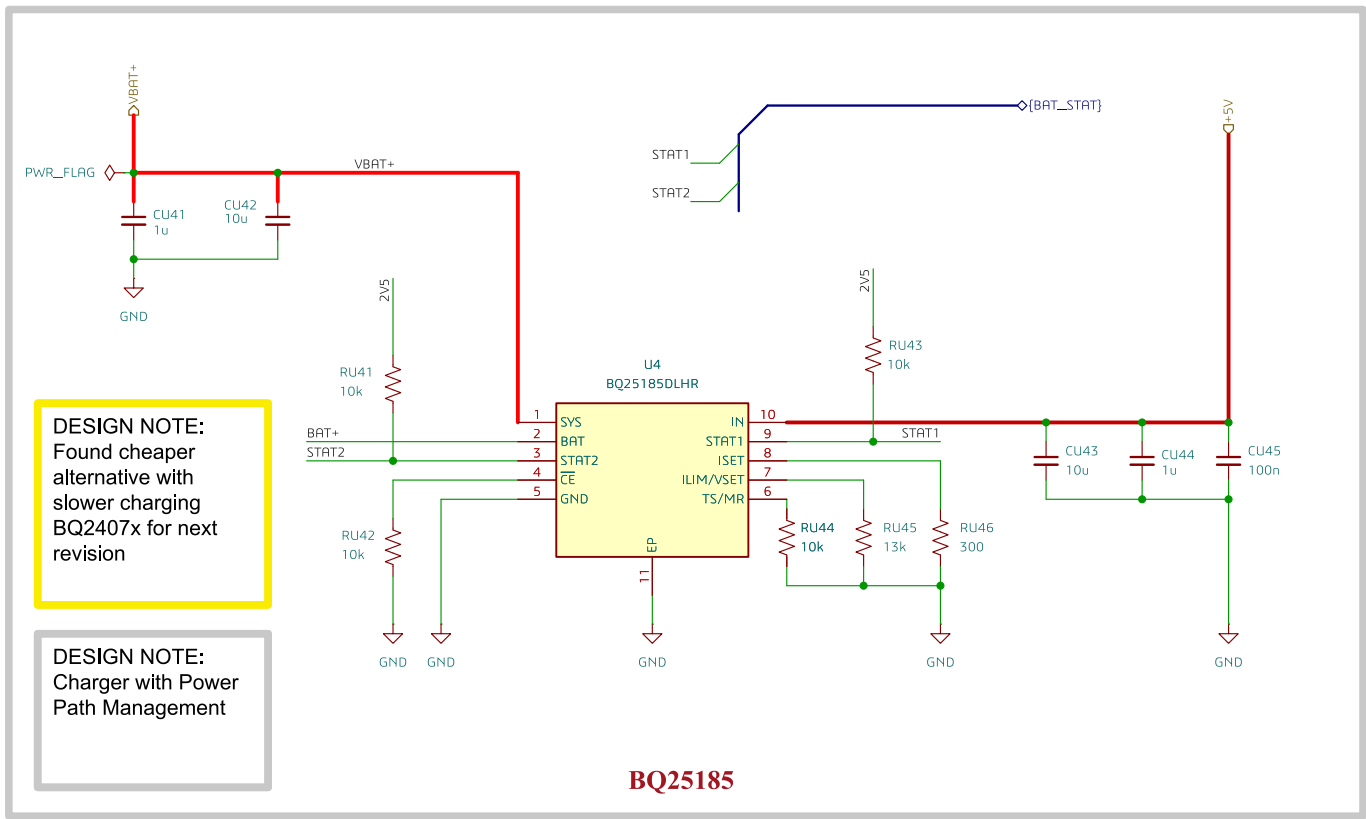
[illegible]

Diagram showing two signals, USB_N and USB_P, connected to a USB symbol. The signals are represented by green lines, and the USB symbol is a blue diamond with the text {USB} inside.

USB

	Comments:	Company: Ahmad Taka Personal		Variant: Variant Name		
		Board Name: STM32F070x6		Project Name: Flashlight Project		
	Sheet Title: POWER - USBC	File Name: power_connector.kicad_sch	Designer: Ahmad Taka	Date: 2025-04-26	Revision: 1.2	
	Sheet Path: /Project Architecture/POWER - USBC/		Reviewer: Travis Ziegler	Size: A4	Sheet: 4 of 12	

[5] POWER - GENERATION



Comments: BQ25185 Datasheet	Company: Ahmad Taka Personal		Variant: Variant Name	
	Board Name: STM32F070x6		Project Name: Flashlight Project	
Sheet Title: POWER - GENERATION	File Name: power_generation.kicad_sch	Designer: Ahmad Taka	Date: 2025-04-26	Revision: 1.2
Sheet Path: /Project Architecture/POWER - GENERATION/		Reviewer: Travis Ziegler	Size: A4	Sheet: 5 of 12

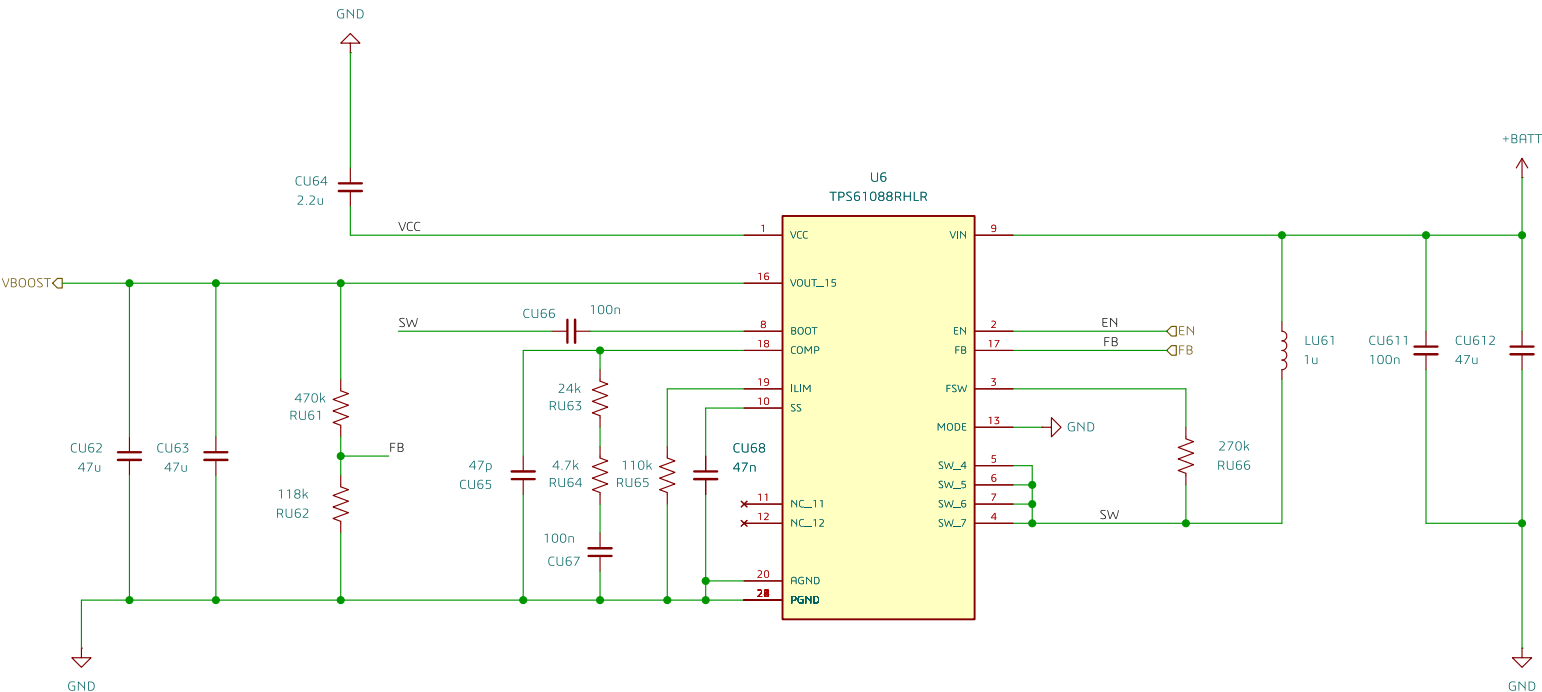
[6] POWER - TPS61088

DESIGN NOTE:
Example text for
cautionary design
notes.

DESIGN NOTE:
Example text for
critical design
notes.

LAYOUT NOTE:
Example text for
critical layout
guidelines.

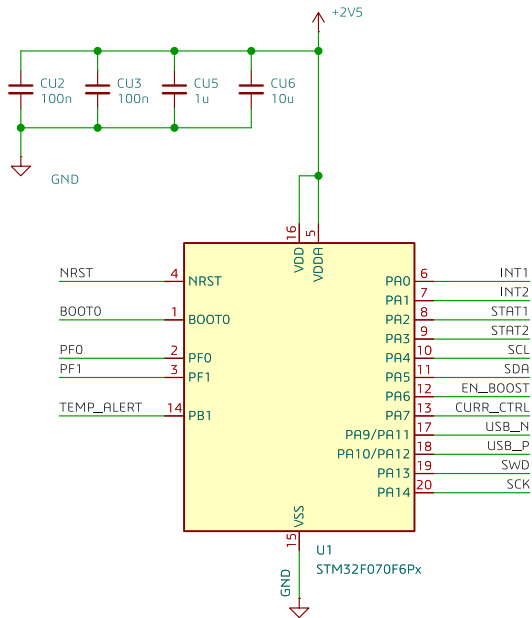
DESIGN NOTE:
Example text for
informational design
notes.



TPS61088 BOOST 3A 6V

	Comments: TPS61088 Datasheet Power Bench Design Example Design		Company: Ahmad Taka Personal		Variant: Variant Name	
			Board Name: STM32F070x6		Project Name: Flashlight Project	
	Sheet Title: POWER - TPS61088		File Name: tps61088.kicad_sch	Designer: Ahmad Taka	Date: 2025-04-26	Revision: 1.2
	Sheet Path: /Project Architecture/POWER - BOOST/			Reviewer: Travis Ziegler	Size: A4	Sheet: 6 of 12

[7] MCU

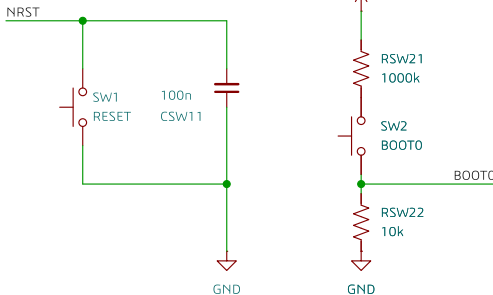


DESIGN NOTE:
We are using software
I2C because the USB
pins take up Hardware
I2C

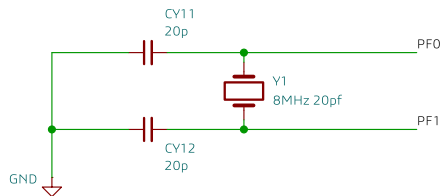
DESIGN NOTE:
USB required for USB
DFU (see AN2606)

LAYOUT NOTE:
Place bypass caps close to
chip

Block Description



RESET AND BOOT



8MHZ OSCILLATOR



BATTERY STATUS



SWD



TEMP ALERT



ACCELERATOR



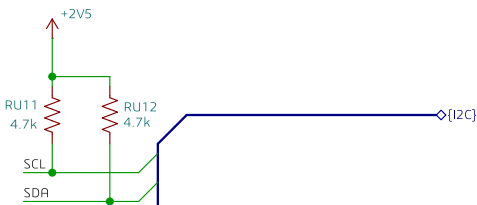
USB



CURRENT CONTROL

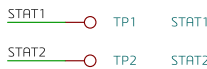


BOOST ENABLE

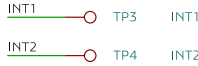


I2C

BATTERY STATUS



ACCELEROMETER



BOOST



TEMP ALERT



CURRENT CONTROL



TESTPOINTS

Comments: AN2606 Design with STM32F070x6 STM32F070x6 Datasheet	Company: Ahmad Taka Personal		Variant: Variant Name	
	Board Name: STM32F070x6		Project Name: Flashlight Project	
	Sheet Title: MCU	File Name: mcu.kicad_sch	Designer: Ahmad Taka	Date: 2025-04-26
Sheet Path: /Project Architecture/MCU/		Reviewer: Travis Ziegler	Size: A3	Revision: 1.2
			Sheet: 7 of 12	

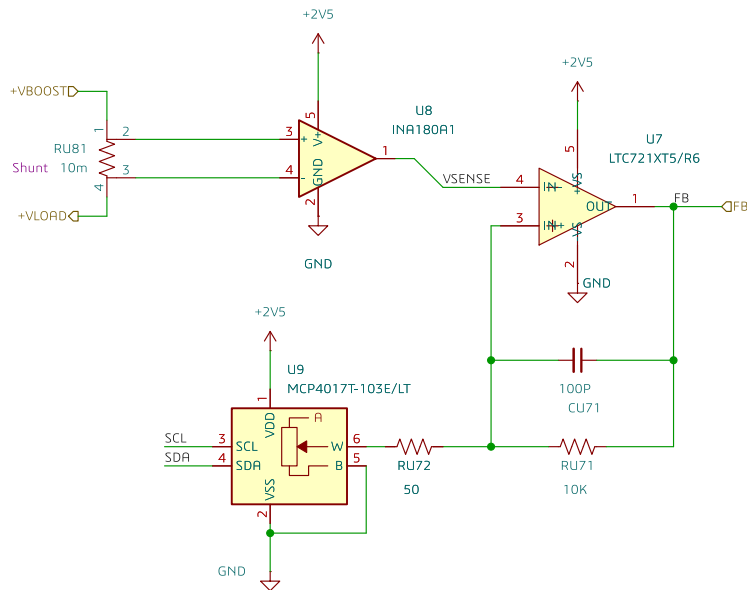
[8] SENSOR - CURRENT

DESIGN NOTE:
Pullups for I2C are
on MCU page

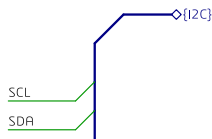
TABLE 5-2: DEVICE I²C ADDRESS

Device	I ² C Address	Comment
MCP4017	'0101111'	
MCP4018	'0101111'	
MCP4019	'0101111'	

LAYOUT NOTE:
Make sure to
check datasheet
for orientation for
silkscreen



HIGH SIDE CURRENT SEMSE



I2C



TESTPOINTS

Comments:

Feedback Circuit
MCP4017T-103E/LT Datasheet

Sheet Title:

SENSOR - CURRENT

Sheet Path:

/Project Architecture/SENSOR - CURRENT/

Company:

Ahmad Taka Personal

Board Name:

STM32F070x6

File Name:

current_sense.kicad_sch

Designer:

Ahmad Taka

Reviewer:

Travis Ziegler

Variant:

Variant Name

Project Name:

Flashlight Project

Date:

2025-04-26

Size:

A4

Revision:

1.2

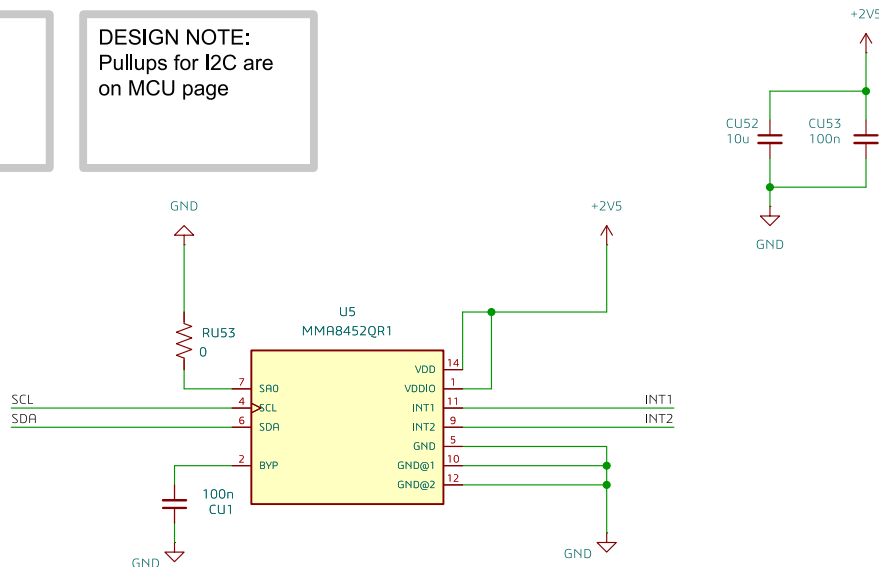
Sheet:

8 of 12

[9] SENSOR - ACCELEROMETER

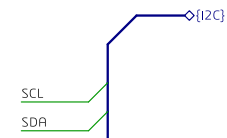
DESIGN NOTE:
ADDR \$1C

DESIGN NOTE:
Pullups for I2C are
on MCU page

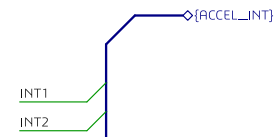


LAYOUT NOTE:
Make sure to
check datasheet
for orientation for
silkscreen

MMA8452QR1 ACCELEROMETER



I2C



INT

Comments:

MMA8452QR1 Datasheet

Company:

Ahmad Taka Personal

Board Name:

STM32F070x6

Variant:

Variant Name

Project Name:

Flashlight Project

Sheet Title:

SENSOR - ACCELEROMETER

File Name:

accelerometer.kicad_sch

Designer:

Ahmad Taka

Date:

2025-04-26

Revision:

1.2

Sheet Path:

/Project Architecture/SENSOR - ACCELERATION/

Reviewer:

Travis Ziegler

Size:

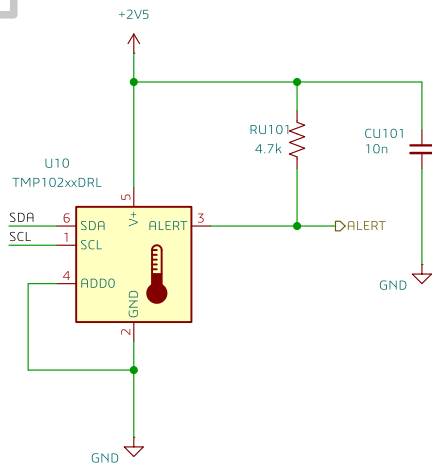
A4

Sheet:

9 of 12

[10] SENSOR - TEMPERATURE

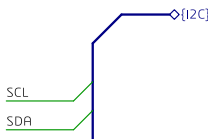
Table 6-4. Address Pin and Target Addresses	
DEVICE TWO-WIRE ADDRESS	A0 PIN CONNECTION
1001000	Ground
1001001	V+
1001010	SDA
1001011	SCL



DESIGN NOTE:
Pullups for I2C are
on MCU page

LAYOUT NOTE:
Make sure to
check datasheet
for orientation for
silkscreen

TMP102 TEMPERATURE SENSOR



I2C

Comments:

TMP102 Datasheet

Company:

Ahmad Taka Personal

Board Name:

STM32F070x6

Variant:

Variant Name

Project Name:

Flashlight Project

Sheet Title:

SENSOR - TEMPERATURE

File Name:

temperature.kicad_sch

Designer:

Ahmad Taka

Date:

2025-04-26

Revision:

1.2

Sheet Path:

/Project Architecture/SENSOR - TEMPERATURE/

Reviewer:

Travis Ziegler

Size:

A4

Sheet:

10 of 12